

AI_LABOR Intelligence Report

CONDITIONALLY_READY

report_id=report_intel_dbab7b01963396b5_v2 version=2 view=PUBLIC

CURRENT STATE

US labor force participation rate (World Bank SL.TLF.ACTI.ZS) ranged 71.58-73.02% over 2010-2022. Trend classification: STABLE. No evidence yet connects this aggregate movement specifically to AI adoption -- a general participation-rate change is not treated as an AI effect. [O-2 round 4 update] Real evidence has since been added beyond the World Bank participation-rate series: (1) OECD AI Exposure Measure methodology -- exposure/capability-gap only, explicitly NOT displacement; (2) US BLS JOLTS Aug 2026 aggregate labor-flow data, little-changed month-over-month, no AI attribution in the release; (3) NBER w31161 field experiment: generative-AI assistant raised customer-support productivity 14% on average (34% for novices), a firm-level productivity finding, NOT employment/headcount evidence; (4) Stanford SIEPR 'Canaries in the Coal Mine': 13% relative employment decline for early-career (22-25) US workers in high-AI-exposure occupations since generative AI adoption -- direct observed employment-effect evidence, but scoped narrowly to that cohort; (5) KDI Korea macro FORECAST (not observed): +3.5% TFP over 10 years, ~256,000 jobs/year displaced; (6) ETLA/Finland peer-reviewed population-level study found NO statistically significant wage/employment divergence by AI exposure in the 2 years after ChatGPT's launch -- genuine counterevidence to a broad displacement reading; (7) Fed FEDS Note: US firm/worker AI-adoption rates (18% of firms by end-2025; 41% of workforce using generative AI at work; adoption only, not a productivity or employment-effect measurement). Four real Hypotheses (HL1-HL4) were constructed this round from this evidence base via the canonical [internal reference withheld in Public view] write path: HL1 (US AI adoption is measurably increasing) = PARTIALLY_SUPPORTED (single Fed synthesis source, source-tier not yet classified -> UNKNOWN-tier guard fires); HL2 (direct observed employment effect concentrated in early-career US workers) = CONTESTED (Stanford SIEPR support vs. ETLA/Finland null-result contradiction -- a genuine geography/population tension, surfaced not resolved, with the OECD exposure!=displacement framing recorded as a generalization guard); HL3 (measurable productivity effect scoped strictly to generative-AI customer support) = PARTIALLY_SUPPORTED (same UNKNOWN-source-tier guard as HL1); HL4 (AI is

driving large-scale, economy-wide job displacement beyond the early-career cohort) = WEAKENED (KDI's 10-year forecast is not yet corroborated, and is actively weakened, by observed aggregate US BLS data showing no AI-attributed change and by Finland's observed null result -- a forecast-vs-observed divergence, not a rejection, since the forecast horizon has not elapsed and geography differs).

KEY SIGNAL

EVIDENCE_INSUFFICIENT

EVIDENCE_INSUFFICIENT

EVIDENCE STATUS

CONDITIONALLY_READY

MAJOR UNCERTAINTY

aggregate participation rate cannot be decomposed by AI-exposure occupation/sector from this indicator alone (ATTRIBUTION_UNCERTAINTY)

WHAT TO WATCH

- ATTRIBUTION_GAP
- COUNTEREVIDENCE_GAP
- POLICY_RESEARCH_GAP
- AI_DISPLACEMENT_VS_PRODUCTIVITY_DISAGGREGATION_GAP

핵심 질문

AVAILABLE

- is AI adoption measurably shifting US labor force participation?

현재 상태

AVAILABLE

- US labor force participation rate (World Bank SL.TLF.ACTI.ZS) ranged 71.58-73.02% over 2010-2022. Trend classification: STABLE. No evidence yet connects this aggregate movement specifically to AI adoption -- a general participation-rate change is not treated as an AI effect. [O-2 round 4 update] Real evidence has since been added beyond the World Bank participation-rate series: (1) OECD AI Exposure Measure methodology -- exposure/capability-gap only, explicitly NOT displacement; (2) US BLS JOLTS Aug 2026 aggregate labor-flow data, little-changed month-over-month, no AI attribution in the release; (3) NBER w31161 field experiment: generative-AI assistant raised customer-support productivity 14% on average (34% for novices), a firm-level productivity finding, NOT employment/headcount evidence; (4) Stanford SIEPR 'Canaries in the Coal Mine': 13% relative employment decline for early-career (22-25) US workers in high-AI-exposure occupations since generative AI adoption -- direct observed employment-effect evidence, but scoped narrowly to that cohort; (5) KDI Korea macro FORECAST (not observed): +3.5% TFP over 10 years, ~256,000 jobs/year displaced; (6) ETLA/Finland peer-reviewed population-level study found NO statistically significant wage/employment divergence by AI exposure in the 2 years after ChatGPT's launch -- genuine counterevidence to a broad displacement reading; (7) Fed FEDS Note: US firm/worker AI-adoption rates (18% of firms by end-2025; 41% of workforce using generative AI at work; adoption only, not a productivity or employment-effect measurement). Four real Hypotheses (HL1-HL4) were constructed this round from this evidence base via the canonical [internal reference withheld in Public view] write path: HL1 (US AI adoption is measurably increasing) = PARTIALLY_SUPPORTED (single Fed synthesis source, source-tier not yet classified -> UNKNOWN-tier guard fires); HL2 (direct observed employment effect concentrated in early-career US workers) = CONTESTED (Stanford SIEPR support vs. ETLA/Finland null-result contradiction -- a genuine geography/population tension, surfaced not resolved, with the OECD exposure!=displacement framing recorded as a generalization guard); HL3 (measurable productivity effect scoped strictly to generative-AI customer support) = PARTIALLY_SUPPORTED (same UNKNOWN-source-tier guard as HL1); HL4 (AI is driving large-scale, economy-wide job displacement beyond the early-career cohort) = WEAKENED (KDI's 10-year forecast is not yet corroborated, and is actively weakened, by observed aggregate US BLS data showing no AI-attributed change and by Finland's observed null result -- a forecast-vs-observed divergence, not a rejection, since the forecast horizon has not elapsed and geography differs).

핵심 주장

AVAILABLE

- 현재 확인되지 않았다. OPEN
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- 현재 확인되지 않았다. OPEN
- 현재 증거만으로 판단하기 어렵다. OPEN
- 현재 확인되지 않았다. OPEN
- 현재 확인되지 않았다. OPEN
- 현재 확인되지 않았다. OPEN
- 현재 확인되지 않았다. OPEN

확인된 사실

INSUFFICIENT_EVIDENCE

Insufficient evidence in the current search scope.

아직 모르는 것 AVAILABLE

- **ATTRIBUTION_GAP**: no sector/occupation-level decomposition linking the original aggregate labor-force-participation indicator to AI adoption specifically -- still true after this round's additions, which rest on separate, newer claims rather than decomposing this indicator
- **COUNTEREVIDENCE_GAP**: PARTIALLY_RESOLVED this round -- ETLA/Finland null-result and BLS JOLTS aggregate-no-change evidence now connected as real counterevidence to a broad AI-displacement reading; still no counterevidence specifically targeting the narrow Stanford SIEPR early-career finding itself (e.g. no replication attempt or alternative explanation study of that specific cohort/occupation set)
- **POLICY_RESEARCH_GAP**: carried over -- still NOT_READY
- **AI_DISPLACEMENT_VS_PRODUCTIVITY_DISAGGREGATION_GAP**: across all 7 new claims, only NBER w31161 (productivity) and Stanford SIEPR (narrow-cohort employment) give any causal/attribution-level evidence; OECD, BLS, KDI, ETLA, and Fed sources are each explicitly scoped to methodology, aggregate-no-attribution, forecast, null-result, or adoption-only respectively -- no source in this corpus yet measures an economy-wide, AI-attributed net employment effect for the general working-age population in any single geography.
- aggregate participation rate cannot be decomposed by AI-exposure occupation/sector from this indicator alone (ATTRIBUTION_UNCERTAINTY)
- none of the 7 new AI_LABOR claims carry an `evidence\_independence` field yet (unlike several AI_ENERGY_INFRA claims), so evidence_evaluation.classify_source_tier() resolves them all to UNKNOWN tier; this is why HL1 and HL3 compute to PARTIALLY_SUPPORTED rather than SUPPORTED despite having a single, uncontradicted, real source each -- the guard is doing its job (source quality was never actually checked), not reflecting a weakness in the underlying evidence itself. (SOURCE_TIER_CLASSIFICATION_GAP)
- four of the round-3 claims (Stanford SIEPR, KDI, ETLA, Fed FEDS Note) share the generic created_from label 'O-2 round 3 WebFetch research' rather than a document-specific identifier (unlike the AI_ENERGY_INFRA corpus's per-document created_from values); this means evidence_evaluation._origin_family_key() would treat them as members of one shared origin family if they were ever combined as multiple supporting items on a single hypothesis in a future round, which could spuriously trip the Derivative Evidence Guard despite being four genuinely independent documents. Not an issue for the 4 hypotheses built this round (each only places at most one of these as supporting evidence), but flagged for a future round before combining them. (SHARED_CREATED_FROM_LABEL)

통계

AVAILABLE

Statistical series: SL.TLF.ACTI.ZS (USA, 2010-2022) -- chart type LINE_CHART_NO_FORECAST, no forecast line, no AI-attribution overlay.

- | | |
|-----------|-------------------|
| Indicator | SL.TLF.ACTI.ZS |
| Geography | USA |
| Period | 2010-2022 |
| Source | src_worldbank_api |
| Trend | STABLE |

실제 관측값 범위를 그대로 사용 -- 추세를 과장하는 축 조정 없음

연구 근거

INSUFFICIENT_EVIDENCE

Insufficient evidence in the current search scope.

정책/규제

INSUFFICIENT_EVIDENCE

Insufficient evidence in the current search scope.

반증

AVAILABLE

- null-result / no-significant-divergence

EVIDENCE_FOUND

 : Kauhanen & Rouvinen (ETLA, Finland, Applied Economics Letters 2025) population-level wage/employment data, 2 years post-ChatGPT
- aggregate-labor-market-no-change

EVIDENCE_FOUND

 : US BLS JOLTS Aug 2026 release -- openings/hires/separations each little changed, no AI attribution in release

대안 설명

AVAILABLE

- General Post 2022 Tech Sector Hiring Slowdown
(GENERAL_POST_2022_TECH_SECTOR_HIRING_SLOWDOWN)
- Differing Labor Market Institutions Us Vs Finland
(DIFFERING_LABOR_MARKET_INSTITUTIONS_US_VS_FINLAND)
- Exposure Not Yet Realized As Displacement (Oecd Framing)
(EXPOSURE_NOT_YET_REALIZED_AS_DISPLACEMENT (OECD framing))
- Forecast Horizon Not Yet Elapsed (Kdi 10-Year Projection)
(FORECAST_HORIZON_NOT_YET_ELAPSED (KDI 10-year projection))

불확실성

AVAILABLE

- aggregate participation rate cannot be decomposed by AI-exposure occupation/sector from this indicator alone (ATTRIBUTION_UNCERTAINTY)
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지리적 맥락

AVAILABLE

- USA (statistic geography field (World Bank country code)): event_country=UNKNOWN
- OECD_CROSS_COUNTRY (OECD AI Exposure Measure methodology paper): cross-country exposure/capability-gap methodology, not a per-country displacement measurement
- USA (BLS JOLTS, NBER w31161, Stanford SIEPR, Fed FEDS Note): US-specific adoption, productivity (customer-support only), and early-career-cohort employment evidence -- not generalized to the full US labor market
- KR (KDI macroeconomic forecast): South Korea 10-year FORECAST only, not an observed outcome; never merged with US/Finland observed data
- FI (Kauhanen & Rouvinen / ETLA population-level study): Finland national wage-earner population data, 2 years post-ChatGPT; a genuine counterevidence data point to a broad early-career-displacement generalization, not a refutation of the narrower US Stanford SIEPR finding

시간적 맥락

AVAILABLE

- 2010 - 2022 (internal data series/event reference (id withheld in Public view))
- 2022-11 - 2026-09 (claims: OECD/BLS/NBER/Stanford-SIEPR/ETLA/Fed-FEDS (see key_claims)) -- post-ChatGPT (Nov 2022) observed-evidence window; kept explicitly separate from the KDI 2026-2036 FORECAST horizon, which must never be read as already observed

근거 지도

AVAILABLE

- Statistic reference (internal id withheld in Public view)
- Evidence Relation reference (internal id withheld in Public view)
- Claim reference (internal id withheld in Public view)
- Claim reference (internal id withheld in Public view)
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- Claim reference (internal id withheld in Public view)
- Claim reference (internal id withheld in Public view)
- Intelligence Object reference (internal id withheld in Public view)

출처

AVAILABLE

- https://api.worldbank.org/v2/country/USA/indicator/SL.TLF.ACTI.ZS?date=2010:2022&format=json&per_page=100 -- non-document provenance reference
- https://www.oecd.org/en/publications/the-oecd-ai-exposure-measure_f3da0f0a-en.html -- non-document provenance reference
- <https://www.bls.gov/news.release/jolts.nr0.htm> -- non-document provenance reference
- <https://www.nber.org/papers/w31161> -- non-document provenance reference
- <https://siepr.stanford.edu/publications/working-paper/canaries-coal-mine-six-facts-about-recent-employment-effects-artificial> -- non-document provenance reference
- https://mobile.newsis.com/view/NISX20260722_0003718118 -- non-document provenance reference
- <https://www.etla.fi/julkaisut/akateemiset-julkaisut/assessing-early-labour-market-effects-of-generative-ai-evidence-from-population-data.md> -- non-document provenance reference
- <https://www.federalreserve.gov/econres/notes/feds-notes/monitoring-ai-adoption-in-the-u-s-economy-20260403.html> -- non-document provenance reference

Sources

1. ^[1] https://api.worldbank.org/v2/country/USA/indicator/SL.TLF.ACTI.ZS?date=2010:2022&format=json&per_page=100 -- NON_DOCUMENT_PROVENANCE (UNKNOWN) [UNKNOWN/UNKNOWN]
2. ^[2] https://www.oecd.org/en/publications/the-oecd-ai-exposure-measure_f3da0f0a-en.html -- NON_DOCUMENT_PROVENANCE (UNKNOWN) [UNKNOWN/UNKNOWN]
3. ^[3] <https://www.bls.gov/news.release/jolts.nr0.htm> -- NON_DOCUMENT_PROVENANCE (UNKNOWN) [UNKNOWN/UNKNOWN]
4. ^[4] <https://www.nber.org/papers/w31161> -- NON_DOCUMENT_PROVENANCE (UNKNOWN) [UNKNOWN/UNKNOWN]
5. ^[5] <https://siepr.stanford.edu/publications/working-paper/canaries-coal-mine-six-facts-about-recent-employment-effects-artificial> -- NON_DOCUMENT_PROVENANCE (UNKNOWN) [UNKNOWN/UNKNOWN]
6. ^[6] https://mobile.news18.com/view/NISX20260722_0003718118 -- NON_DOCUMENT_PROVENANCE (UNKNOWN) [UNKNOWN/UNKNOWN]
7. ^[7] <https://www.etla.fi/julkaisut/akateemiset-julkaisut/assessing-early-labour-market-effects-of-generative-ai-evidence-from-population-data.md> -- NON_DOCUMENT_PROVENANCE (UNKNOWN) [UNKNOWN/UNKNOWN]
8. ^[8] <https://www.federalreserve.gov/econres/notes/feds-notes/monitoring-ai-adoption-in-the-u-s-economy-20260403.html> -- NON_DOCUMENT_PROVENANCE (UNKNOWN) [UNKNOWN/UNKNOWN]

METAXIS Intelligence Observatory -- structured evidence, not editorial conclusions.